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Prediction of stand quality characteristics in sweet chestnut forests in NW Spain by combining terrain attributes, spectral textural features and landscape metrics

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ABSTRACT

Chestnut woods are outstanding elements of European cultural landscapes, and are also recognised as habitats of interest in the Natura 2000 European network of protected areas. Despite their importance, they are threatened by land abandonment and recent rural landscape changes. The loss of *Castanea sativa* cover and its dominance as a characteristic species may indicate forest degradation, which can occur as a result of natural succession dynamics or invasion by alien species.

In this study we aimed to model the conservation status of chestnut stands in the NW Iberian peninsula in terms of their environmental and cultural value, formulated as a quality index that considers the maturity and purity of forest stands. We used features available at wide geographic scales as potential predictors, namely topographic parameters derived from Digital Elevation Models, land-cover heterogeneity determined by means of satellite image texture and landscape metrics. According to the results, altitude emerged as a major factor determining chestnut stand quality, marking a clear difference between highland and lowland environments. The combination of terrain, texture and patch shape refined the classification of each of these scenarios. We discuss the significance of these predictors as environmental controls and indicators of anthropogenic effects.

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1. Introduction

Sweet chestnut (*Castanea sativa* Mill.) woodland covers more than 2.5 million hectares in Europe, with a distribution area ranging from the Southern Mediterranean to Central, Atlantic and Eastern Europe (Conedera et al., 2004a). In NW Spain, chestnut occurs under a wide variety of climates, ranging from Mediterranean conditions with an annual rainfall of 600–700 mm and marked summer drought, to Atlantic climates with more than 1700 mm of rain distributed evenly throughout the year. Since this species is sensitive to severe cold and frost, and fruit production depends on a certain degree of warmth and insolation in the summer, chestnut stands seldom occur at altitudes above 1200 m (Diaz Varela et al., 2009). The types of chestnut woodland existing at present differ widely in terms of stand structure and main productive aim, and include plantations for nut (and sometimes wood) production, coppice stands and high forest, mainly planted for timber (Bourgeois et al., 2004).

Sweet chestnut is a native species in the NW Iberian Peninsula (Conedera et al., 2004b; Krebs et al., 2004). Pollen records indicate the presence of *Castanea sativa*, together with other thermophillous species during the Late Glacial period in NW Spain, and suggest the presence of sheltered areas where these species found refugia during the Würm glaciation, spreading out throughout the Holocene as part of mesophyllous forests (Ramil Rego et al., 1998). As in other parts of Europe, systematic cultivation of the species for fruit and wood production took place during the Middle Ages (Conedera et al., 2004a), probably promoted by monastic orders. In fact, within the framework of the medieval privilege system, there are frequent historical records from monasteries referring to land contracts that involved planting either chestnut forest or individual trees scattered within cultivated land (Ríos Rodriguez, 1997). Moreover, due to environmental constraints in mountainous areas, sweet chestnut has for many centuries been essential to the human diet, while leaves and fruits in the understory were eaten by livestock.

As in other regions, sweet chestnut woods have traditionally been managed in NW Spain by two different regimes (Conedera et al., 2001): coppice or pure chestnut forests, regenerated from

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adventitious or dormant buds in order to produce poles and firewood, and orchards and grafted chestnut in open stands, to produce mainly fruit as well as pasture, forage, litter and timber. In some areas of NW Spain the understory of these orchards was completely removed, ploughed and sown with wheat.

During the 17th and 18th centuries, chestnut stands decreased in coastal areas, probably as a consequence of the high demand for wood, overgrazing and the expansion of more productive forest and crop species. Chestnut stands have decreased dramatically in extension since the end of the 18th and beginning of the 19th century, particularly at low elevations, because of the spread of ink disease (*Phytophthora cambivora* and *P. cinnamomi*), which has caused extensive damage to chestnut trees in Atlantic areas, and more recently because of the effects of chestnut blight (*Cryphonectria parasitica*) (Bouhier, 1979; Mansilla Vázquez et al., 2000). In recent decades, chestnut stands in NW Spain have declined significantly as a result of changes in the farming system, involving the introduction of new agricultural crops, abandonment of traditional cultural practices and their replacement by pine or eucalyptus afforestations (Bouhier, 1979; Diaz Varela et al., 2009).

Chestnut woods have been recognised as habitats of interest in the European Natura 2000 network and also acknowledged as characteristic cultural landscapes of Mediterranean and Atlantic regions of Europe. Despite the environmental and cultural interest in chestnut stands, they are threatened by a series of factors such as pathogenic fungi, the abandonment of traditional management and recent rural landscape changes in both Atlantic and Mediterranean Europe (Cevasco, 2009; Diaz Varela et al., 2009). In fact, the conservation status of chestnut stands is regarded as "not favourable" in Alpine, Continental and Mediterranean regions, while information is lacking for the Atlantic region, according to the European Environment Agency (2009). One of the main threats to the long term conservation of this habitat is the natural evolution of chestnut stands towards some kind of native forest due to natural succession dynamics or invasion by alien woodland species.

Taking these threats into account, it is necessary to design simple, repeatable and spatially explicit methods for large scale assessment of chestnut forest conservation status in the European Union. In this study we explore the potential of an automatic and spatially exhaustive assessment of the conservation status of chestnut woodlands, using available large scale datasets including remote sensed imagery, forest inventory maps and digital terrain models. We hypothesized that some simple stand variables, such as forest structure and composition, can be used as reliable indicators of the conservation status of chestnut woodlands (Treitz and Howarth, 2000). We used statistical methods to explore the predictive value of several datasets potentially related to these indicators, such as topographic features (Geri et al., 2010; Lindenmayer et al., 1999), patch morphology metrics (Saura and Carballal, 2004) and satellite image texture (Franklin et al., 2001; Kayitakire et al., 2006).

2. Material and methods

2.1. Study area

The study was carried out in an area of the NW Iberian Peninsula where there are good examples of chestnut forest stands in different states of conservation. According to the definition of the European Commission (2005) and the principles of the European Directive 92/43/CE, conservation status results from the sum of the influences acting on a given habitat that may affect its long-term natural distribution, structure and functions as well as the long-term survival of the characteristic species. In the present study, good or "adequate" conservation status was assumed in sites where chestnut stands are prospering (in both quality and extent/population) and with good prospects to do so in future, while a poor or "inadequate" status was assumed in sites where the opposite occurs as a result of competition with other species, land use change, diseases or any other factor.

In order to provide a sufficiently large dataset to include the variability in forest stand conditions and to avoid distortions due to differences in remote sensed imagery dates, we considered an area of approximately 24 448 km^2 corresponding to the extent of the Landsat TM scene 204-30 (Fig. 1).

Altitudes in the study area range from sea level to about 2000 m a.s.l., shaping a contrasted relief. In the western area, low and smooth mountain systems that barely surpass 1000 m a.s.l. (Dorsal Galega, Serras Septentrionais de Galicia) alternate with relatively wide tectonic depressions, plateaux (Terra Cha, Monforte) and valleys following tectonic faults. Towards the east, a succession of mountain massifs marks the transition to the western end of the Cantabrian Range with higher altitudes (2000–2100 m a.s.l.), rough relief and generally narrow and deep valleys, with the exception of the small Bierzo plateau. From an administrative point of view, most of the study area is in the region of Galicia, although parts are in the regions of Asturias and Castilla-Leon (all in Spain).

Biogeographically most of the study area is included in the Atlantic Region, although part of the SE quadrant of the area corresponds to the Mediterranean region. However, the exact extent of the Mediterranean domain in the area is a matter of discussion (European Environment Agency, 2008; Rivas-Martínez, 1987; Rivas-Martinez and Rivas-Saenz, 2009). Natural vegetation is characterised by distinct deciduous forests, mainly dominated by *Quercus robur* and *Quercus petraea*, coexisting towards the west with *Betula alba* and *Fagus sylvatica* and interspersed with mixed mesophytic woodland in the valley bottoms. Transition to continental and Mediterranean environments is characterised by the dominance of *Quercus pyrenaica* forests, while some evergreen or sclerophyllous forests are restricted to drier and warmer locations (Ramil Rego et al., 1998; Rivas-Martínez, 1987, 2007).

The climate in the study area is defined by a transition from oceanic or even hyper-oceanic, with annual rainfall up to 2400 mm, without marked drought periods -because of the Atlantic influence- to sub-Mediterranean and Continental conditions towards inland areas where annual rainfall may be as low as 600 mm and where marked summer drought occurs at certain points in the SE. Average annual temperatures range from around 4–5 °C to more than 14 °C. The temperature is relatively mild in the area, but relatively cold mountain climates occur in the mountain systems, preventing the existence of forests (tree or forest line) from altitudes of 700–1700 m a.s.l. depending on the mountain location (Muñoz Sobrino et al., 2005; Rivas-Martinez and Rivas-Saenz, 2009).

2.2. Datasets

2.2.1. Forest stand variables

Chestnut forest variables were extracted from the Digital Forest Map of Spain at a scale of 1:50 000 (Ministerio de Medio Ambiente, 2002). This map is based on digitalization of forests from satellite images or aerial orthophotographs and supported by fieldwork and ancillary data. The minimum mappable area is 2.5 ha for forest and 6.25 ha for other types of land cover. This map provides information on overall tree species composition (expressed as a percentage of the total), canopy cover by individual species (in 10% intervals) and stage of stand development (in a scale from one to four) based on the criteria proposed by Schütz (1990). The stage of stand development is defined on the basis of the theoretical changes in stand structure, from establishment to mature stages. In this case, stage 1 ranges from the seedlings to the crown closure, stage 2 starts with crown closure and lasts until the loss of lower foliage and branches due to crown shadowing, stage 3 lasts until the trees reach a DBH

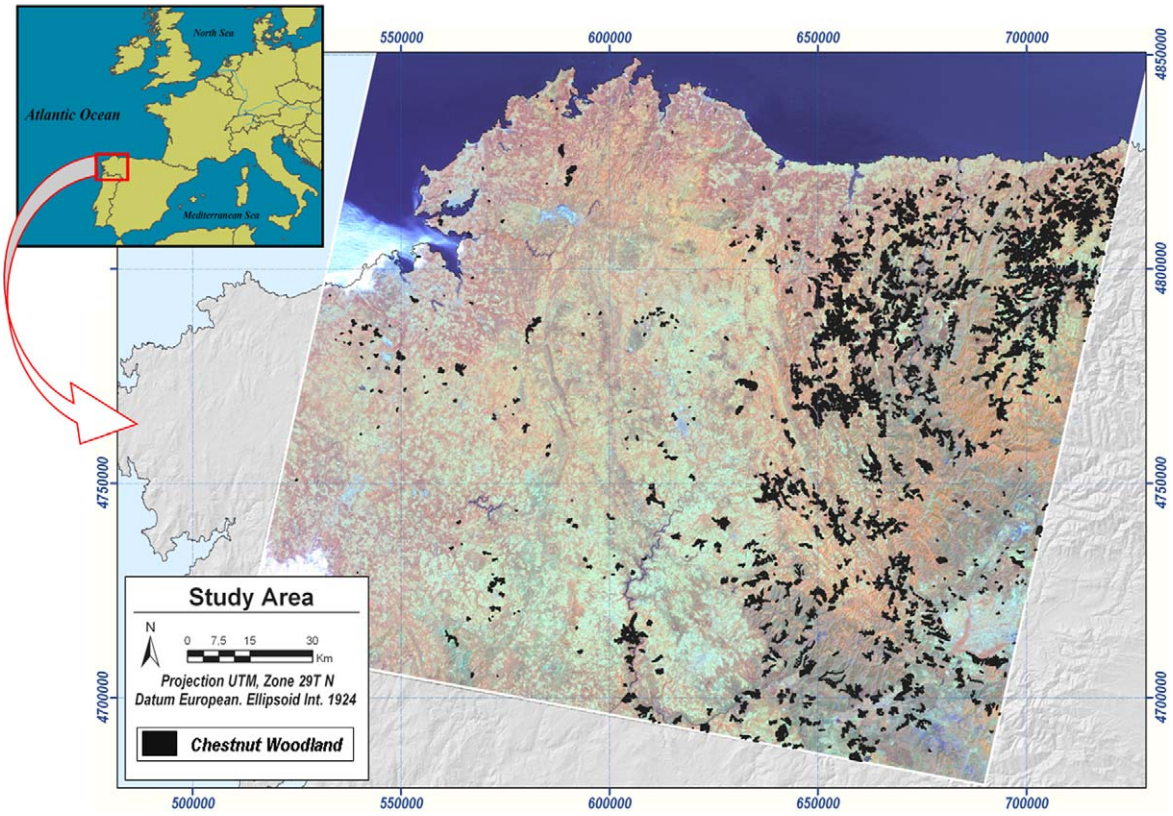


Fig. 1. Study area and distribution of chestnut forest patches.

(Diameter at Breast Height) of 20 cm and stage 4 correspond to mature stands in which DBH is at least over 20 cm.

We queried the map to identify and extract the chestnut forest patches in the study area. Within the framework of the present study, and as we focused on dense forest with a significant share of fully developed chestnut in the canopy, we considered patches as those areas meeting the following three conditions: (i) more than 60% tree species crown cover (ii) more than 20% chestnut crown cover, and (iii) chestnut stage of stand development from 2 to 4. This provided us with a total of 1585 chestnut patches.

In regard to the objective of modelling the chestnut forest condition, assuming a double role for the forest as a cultural landscape and seminatural habitat of conservation importance, we proposed a binary quality index (high and low quality stands) comprised of the purity and maturity of the tree canopy (Table 1). The degree of chestnut stand purity was parameterised in terms of the cover of other species, as we hypothesised that a pure but open formation dominated by chestnut will be better preserved than other more dense stands invaded by other tree species. Maturity was evaluated directly by the stage of chestnut stand development. The quality index takes a value of 1 (low quality class) when the chestnut in the forest stand corresponds to second and third stages of stand development (i.e. stands with crown closure and trees with an average DBH < 20 cm), or trees belonging to the fourth stage of stand development (DBH > 20 cm) and more than 20% of canopy cover by other species. A quality index value of 2 (high quality class) corresponds to forest stands meeting the conditions of fourth stage of stand development and less than 30% of other crown cover by species, which will probably correspond to old growth of traditionally grafted chestnut orchards.

We arbitrarily considered a binary index rather than a quality scale with several intermediate levels because the former is closer to the definition of conservation status given in Section 2.1 (European Commission, 2005; European Directive 92/43/CE). Hence, class 1 stands (low quality stands) can be considered as those chestnut stands with unfavourable conservation status, due to strong competition from other species, and facing uncertain changes depending on the particular environmental conditions, socioeconomic setting and forest stand dynamics. On the other hand, the class 2 stands (high quality stands) roughly correspond to stands with favourable conservation status, where the dominance and maturity of the chestnut individuals in the stand facilitate their persistence and future development.

2.2.2. Morphometric landscape patterns

Two complementary approaches were used to extract a set of morphometric parameters from the selected forest patches. Landscape metrics, or quantitative indices that describe landscape structures or patterns (O'Neill et al., 1988), were combined with Morphological Spatial Pattern Analysis (MSPA), which has been developed for the analysis of forest functional connectivity (Vogt et al., 2009). The landscape indices finally chosen were patch area (MPS), patch edge (TE), shape index (MSI), perimeter area ratio (MPAR), fractal dimension (MFRACT), and number of shape characteristic points (NSCP) (see McGarigal et al., 2002; Moser et al., 2002 for formulae and further explanation). The indices were calculated with V-Late extension for ArcGIS 9 software (Lang and Tiede, 2003), except for NSCP, for which a specific script developed for ArcView 3.3 was used (Moser et al., 2002). The parameters used and their acronyms are shown in Table 2. Calculation was made at patch level, i.e. indices were specifically calculated for each individual patch. MSPA was performed with GUIDOS software (Graphical User Interface for the Description of image Objects and their Shapes, Vogt et al., 2009). The output of the MSPA is a raster layer in which patch cells are assigned to seven morphological spatial pattern classes (edge, core, perforated, islets, bridge, loop and branch), thus

Table 1
Criteria for maturity and density in the definition of quality of chestnut forest stand. Q1 stands for quality 1 (low quality stand); Q2 stands for quality 2 (high quality stand).

		Relative cover (in %) by other species in chestnut stands										
		0	10	20	30	40	50	60	70	80	90	100
Stage of stand development	2nd	Q1	Q1	Q1	Q1	Q1	Q1	Q1	Q1	Q1	Q1	Q1
	3rd	Q1	Q1	Q1	Q1	Q1	Q1	Q1	Q1	Q1	Q1	Q1
	4th	Q2	Q2	Q2	Q1	Q1	Q1	Q1	Q1	Q1	Q1	Q1

defining the spatial arrangement by its functional capabilities. The software was run with the default parameters for the computation of MSPA, considering one pixel (30 m) as the edge width. The patch percentage of each morphological class was computed in order to convert this information to patch level.

2.2.3. Terrain relief attributes

Considering that terrain relief may play a key role in landscape structure and in vegetation pattern in general (Hoechstetter et al., 2008), we selected several terrain features that showed potential as predictors of forest distribution (Lindenmayer et al., 1999). First and second order terrain features were derived from a raster digital elevation model with a spatial resolution of 30 m (Table 2). Some of these terrain features are also widely used in hydrological, geomorphological and ecological studies (Wilson and Gallant, 2000), whereas others address vegetation and forest assessment more specifically (McNab, 1989; Roberts and Cooper, 1989)

2.2.4. Remote sensing textural features

Image texture has shown potential for predicting forest stand characteristics (see e.g. Franklin et al., 2001). Eight texture features were computed on the basis of kernel grey level co-occurrence matrices (Haralick et al., 1973), on the well known greenness index NDVI (Table 2) computed from a Landsat 7 satellite image from 24 July 1999, in order to assess their value as predictors of chestnut stand quality at patch level. The software package PCI™ 9.1 was used, with direction set at 45°, displacement of one pixel and a window size of 5×5 pixels. A relatively small kernel size was established, as some small or narrow patches are foreseen, and also to avoid interference from the neighbouring patches.

2.3. Statistical analysis

Classification and regression trees (CaRT) were used to classify and interpret the internal structure of the dataset in order to assess the potential of the different variables as chestnut patch quality predictors (Moore et al., 1991; Vayssiéres et al., 2000). The objective of CaRT is to classify a set into several groups by application of a rule that displays them in the form of a binary tree (Breiman et al., 1984), which is determined by a procedure known as recursive partitioning. CaRT methods do not have to conform to probability distribution restrictions and there is no assumption of linearity or the need to pre-specify a probability distribution for the errors (Bell, 1999).

Every tree split is described by the value of one predictor, chosen in such a way that all objects in a child group have more similar response variable values. The split for continuous variables is defined by "$x_i < a_j$", where x_i is the selected descriptor or explanatory variable and a_j its split value. To choose the most appropriate values of x_i and a_j, CaRT uses an algorithm in which all descriptors and all split values are considered, selecting those giving the best reduction in impurity between the parent group (t_p) and the child groups (t_L and t_R) (Deconinck et al., 2005). Mathematically this is expressed as (Eq. (1)):

$$\Delta i(s, t_p) = i(t_p) - p_L i(t_L) - p_R i(t_R) \quad (1)$$

Table 2
Independent variables in the CaRT.

Variable group	Variable	Acronym	Reference
Landscape pattern	Patch area (m^2)	MPS	
	Patch edge (m)	TE	
	Shape index	MSI	Lang and Tiede (2003)
	Perimeter area ratio	MPAR	
	Fractal dimension	MFRACT	
	Number of shape characteristic points	NSCP	Moser et al. (2002)
	Fraction of edge	EDGE	
	Fraction of core	CORE	
	Fraction of perforated	PERFOR	
	Fraction of islets	ISLET	Vogt et al. (2009)
	Fraction of bridge	BRIDGE	
	Fraction of loop	LOOP	
	Fraction of branch	BRANCH	
Terrain	Elevation (m a.s.l.)	DEM	
	Curvature	CURV	
	Plan curvature	PLCUR	Wilson and Gallant (2000)
	Profile curvature	PROCUR	
	Slope (degrees)	SLOPE	
	Transformed aspect	TRASP	Roberts and Cooper (1989)
	Terrain shape index	TERSHP	McNab (1989)
Image texture	Contrast	CONTR	
	Correlation	CORR	
	Dissimilarity	DISSIM	
	Entropy	ENTROP	Harlick (1973)
	Homogeneity	HOMOG	
	Mean	MEAN	
	Second moment	SECMOM	
	Standard deviation	STNDE	

where i is the impurity, s the candidate split value, and p_L and p_R are the fractions of the objects in the left and right child groups, respectively.

The impurity is defined as the total sum of squares of the deviations of the individual responses from the mean response of the group, computed by the following expression (Eq. (2)):

$$i(t) = \sum_n (y_n - \bar{y}(t))^2 \quad (2)$$

where $i(t)$ is the impurity of group t, y_n the value of the response variable for object x_n and $\bar{y}(t)$ the mean of the response variable in group t.

This process is repeated for each child group until the maximal tree is grown. The cost-complexity procedure is used to simplify trees, and allows pruning and reduction of the nodes on a tree, to produce smaller sub-trees. Comparison of sub-trees is based on a cost-complexity measure, $r\alpha(T)$, which is calculated for a given sub-tree (T) by Eq. (3) (LeBlanc and Tibshirani, 1998):

$$r_\alpha(T) = \sum_{t \in \tilde{T}} \sum_n (y_n - \bar{y}(t))^2 + \alpha \cdot \left|\tilde{T}\right| \quad (3)$$

where $\tilde{T}$ are terminal nodes, $\left|\tilde{T}\right|$ is the number of terminal nodes in T and α is the cost-complexity parameter. For any value of α, the cost-complexity pruning algorithm can efficiently produce the subtree T' that minimizes $r\alpha(T)$. Finally, the predictive error was calculated as the root mean square deviations between observed and predicted values.

We included 43 variables in the Classification and Regression Tree (CaRT) procedure, using the statistical package SPSS™ 16.0. As terrain relief and texture information are calculated at pixel level, we computed the mean (MN) and standard deviation (SD) of these features for each patch in order to obtain information at patch level. The final dataset comprises 16 texture variables (MN and SD for the eight texture features), 14 terrain variables (MN and SD for the seven terrain features) and 13 patch morphology variables (six patch landscape metrics along with the percentages of each of the seven typologies of the MPSA analysis for a given forest patch). The overall descriptive statistics computed at patch level for all the variables included in the CaRT are shown in Table 3, with the variables finally selected by the model as predictors marked in bold type.

The CaRT analysis was carried out considering a minimum parent node size of 50 and a minimum child size of 25. The resulting classification tree was pruned to avoid model overfitting, by erasing subtrees by the cost-complexity pruning process implemented in the statistical package, considering a maximum difference in risk of one standard error.

In order to assess the performance of the model, we generated a contingency matrix to compare the observed (rows) and predicted (columns) membership in classes. As in this case we were interested in enhancing the model performance, we decided not to split the original dataset into a training and a validation sample and therefore used the whole dataset to produce the model (see e.g. Fielding and Bell, 1997 for a more detailed discussion on the subject). We subsequently computed overall accuracy, commission and omission errors along with the overall and conditional KAPPA statistic as measures of model accuracy (Bishop et al., 1975; Congalton, 1991).

3. Results

Observed and predicted values of chestnut stand quality classification trees are summarised as a contingency matrix in Table 4, along with the classification accuracy statistics. The spatial distribution of the location agreements and disagreements between observations and predictions of the model are shown in Fig. 2.

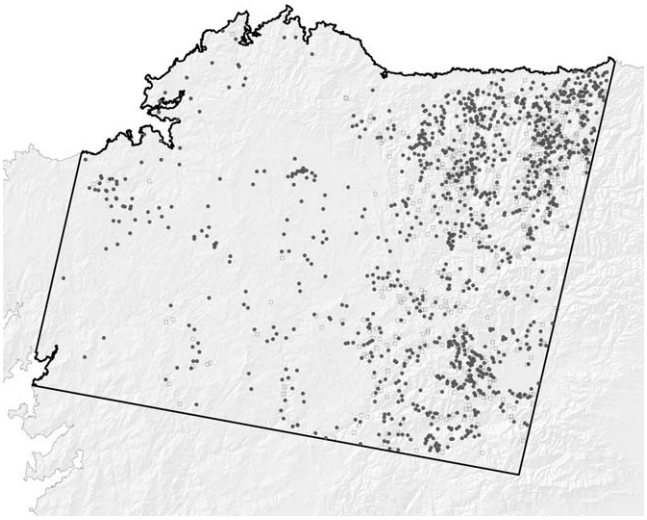


Fig. 2. Spatial distribution of the centroids of chestnut forest patches. Solid circles represent cases of agreement between observed and predicted quality; hollow squares represent disagreement.

The tree classified chestnut forest quality with an overall accuracy of 71%. The kappa coefficient reached a value of 0.38, corresponding to a fair agreement between the observed and classified values according to the qualitative scale of Landis and Koch (1977). Omission errors were slightly lower than commission errors in class 1, indicating a relatively restrictive definition of classification criteria for this class, while this situation is reversed for class 2, indicating that the classification criteria for this class were possibly too broad. Values of per-class conditional Kappa reached a value of 0.40 for class 1 and 0.37 for class 2, in both cases showing fair agreement between the observed and classified values.

The classification tree is shown in Fig. 3, with a structure with five splitting levels and ten terminal nodes. The tree integrates variables belonging to the three types considered (i.e. terrain, image texture and patch morphology). Terrain attributes were by far the most important in the model (Fig. 4), particularly the elevation, which splits the parent node of the tree, along with slope and aspect. Second degree terrain features such as curvature only emerged in the final splits. Spectral texture variables, namely the mean and standard deviation of the texture feature "MEAN" were also included in the intermediate levels of the tree. Only one variable of spatial pattern (mean patch index) was included in the tree, contributing to the discrimination of two terminal nodes in the fifth split level.

Seven out of the ten classification groups belong to class 1 (low quality stands), and show a different degree of purity in their composition, ranging from 64 to 91% of membership in the class. The three groups in class 2 (high quality stands) tended to show a higher degree of mixture, with values of 59, 65 and 67% of membership in this class, respectively.

Use of the classification tree also enabled us to assess the interactions between predictors in order to trace pathways for understanding the ecological meaning of the predicted groups. Hence, the most important predictor (patch elevation, DEM_ME), with a split value at around 600 m a.s.l. established the difference between lowland (valleys and plateaus) and mountains in the study area. Class 1 chestnut forests tend to occur in lowlands while class 2 stands are more frequent above this altitude.

In the lowlands, mean patch slope again marked a clear difference, with an almost pure group of low quality stands occurring on slopes lower than 15 degrees. Patches on steeper slopes were then split on the basis of image texture criteria, for which values of the "Mean" texture feature lower than 45 gave rise to a node with

Table 3
Descriptive statistics of the independent variables originally included in the analyses. In bold type, variables finally selected by the CaRT as predictors.

Variable group	Variable	Mean	Median	Std. Dev.	Minimum	Maximum	Range
Landscape pattern	MPS (ha)	59.718	24.576	107.947	0.150	1304.672	1304.523
	TE	7691.871	4389.870	10,918.053	194.850	147,093.040	146,898.190
	MSI	**2.815**	**2.532**	**1.322**	**1.090**	**13.500**	**12.420**
	MPAR	0.021	0.018	0.012	0.000	0.130	0.130
	MFRACT	1.356	1.355	0.049	1.220	1.540	0.320
	NSCP	111.868	64.000	161.202	3.000	2387.000	2384.000
	EDGE	0.355	0.362	0.119	0.000	1.000	1.000
	CORE	0.413	0.433	0.211	0.000	0.860	0.860
	PERFOR	0.001	0.000	0.004	0.000	0.080	0.080
	ISLET	0.039	0.000	0.182	0.000	1.000	1.000
	BRIDGE	0.037	0.013	0.056	0.000	0.440	0.440
	LOOP	0.005	0.000	0.021	0.000	0.330	0.330
Terrain	BRANCH	0.150	0.102	0.153	0.000	1.000	1.000
	DEM_ME	**552.060**	**556.374**	**209.342**	**3.570**	**1276.680**	**1273.110**
	DEM_SD	55.085	48.615	35.757	0.000	210.830	210.830
	CURV_ME	−0.158	−0.104	0.292	−2.580	1.140	3.720
	CURV_SD	1.047	1.069	0.460	0.000	2.780	2.780
	PLCUR_ME	−0.058	−0.027	0.148	−1.170	0.660	1.820
	PLCUR_SD	0.630	0.647	0.290	0.000	1.740	1.740
	PROCUR_ME	**0.101**	**0.070**	**0.180**	**−0.700**	**1.650**	**2.350**
	PROCUR_SD	**0.582**	**0.565**	**0.273**	**0.000**	**1.810**	**1.810**
	SLOPE_ME	**21.827**	**23.214**	**7.259**	**0.970**	**42.460**	**41.490**
	SLOPE_ST	5.832	5.995	2.012	0.000	16.410	16.410
	TRASP_ME	**0.382**	**0.342**	**0.223**	**0.000**	**0.990**	**0.990**
	TRASP_SD	**0.228**	**0.235**	**0.089**	**0.000**	**0.430**	**0.430**
	TERSHP_ME	0.006	0.004	0.012	−0.050	0.100	0.150
	TERSHP_SD	0.042	0.043	0.018	0.000	0.110	0.110
Image texture	CONTR_ME	89.050	82.174	47.946	0.060	412.790	412.730
	CONTR_SD	71.482	70.892	30.163	0.000	197.220	197.220
	CORR_ME	0.273	0.265	0.101	−0.140	0.870	1.010
	CORR_SD	0.279	0.284	0.036	0.000	0.450	0.450
	DISSIM_ME	6.270	6.275	2.115	0.060	15.390	15.330
	DISSIM_SD	3.014	3.052	0.866	0.000	5.940	5.940
	ENTROP_ME	3.085	3.201	0.327	0.240	3.460	3.220
	ENTROP_SD	0.365	0.285	0.265	0.000	1.310	1.310
	HOMOG_ME	0.254	0.224	0.113	0.050	0.970	0.930
	HOMOG_SD	0.143	0.132	0.063	0.000	0.340	0.340
	MEAN_ME	**46.956**	**48.395**	**8.886**	**10.080**	**62.420**	**52.340**
	MEAN_SD	8.023	8.188	2.725	0.000	16.480	16.480
	SECMOM_ME	0.072	0.049	0.061	0.030	0.870	0.840
	SECMOM_SD	0.059	0.027	0.073	0.000	0.380	0.380
	STNDE_ME	**7.277**	**7.337**	**2.342**	**0.110**	**16.900**	**16.800**
	STNDE_SD	3.327	3.361	0.961	0.000	6.140	6.140

an almost even mixture of the two classes, while values higher than 45 corresponded to a group dominated by the high quality class. Following the pathway from this node, the heterogeneity of along-slope curvature (standard deviation of profile curvature (PROCUR_SD) and general aspect of the patch (TRASP_ME) enabled us to discriminate three new groups. A group of high quality stands corresponded to more uniform (less changes of along-slope curvature) and northern aspect slopes, while two groups dominated by low quality stands corresponded to slopes with a more complicated relief or a combination of more uniform relief and south-facing slopes.

In the highlands, a group of high quality stands was discriminated on the basis of their low variation in aspect within a given patch (TRASP_SD), which could be identified with patches confined to particular locations and not covering all the landscape (i.e. series of slopes with different aspects). The variable mean of profile curvature (PROCURV_ME), related to the rate of change of slope or along slope curvature, enabled discrimination of a low quality group, corresponding to values lower than 0.016, which indicate concave slope profiles (negative curvature) or no slope (curvature 0) (de Smith et al., 2007). The next split in the tree was the mean of the texture feature "standard deviation" (STNDE_ME), for which values lower than 4.9 led to a terminal node of low quality stands. This suggests that patches with relatively low texture values correspond to young and/or mixed chestnut forests in the highlands and under particular physiographic conditions. Finally two groups were split by the variable mean shape index (MSI), whereby relatively compact patch shapes (MSI $\leq$ 4.7) corresponded to low quality stands

Table 4
Contingency table of predicted against observed values used in the classification tree. Rows correspond to the values predicted by the CaRT and columns represent the observed values used in testing the model.

		Observed					
		Class 1	Class 2	Row total	Omission error	Commission error	Conditional Kappa
Predicted	Class 1	753	218	971	24.32	22.45	0.40
	Class 2	242	372	614	36.95	39.41	0.37
	Column Total	995	590	1585			

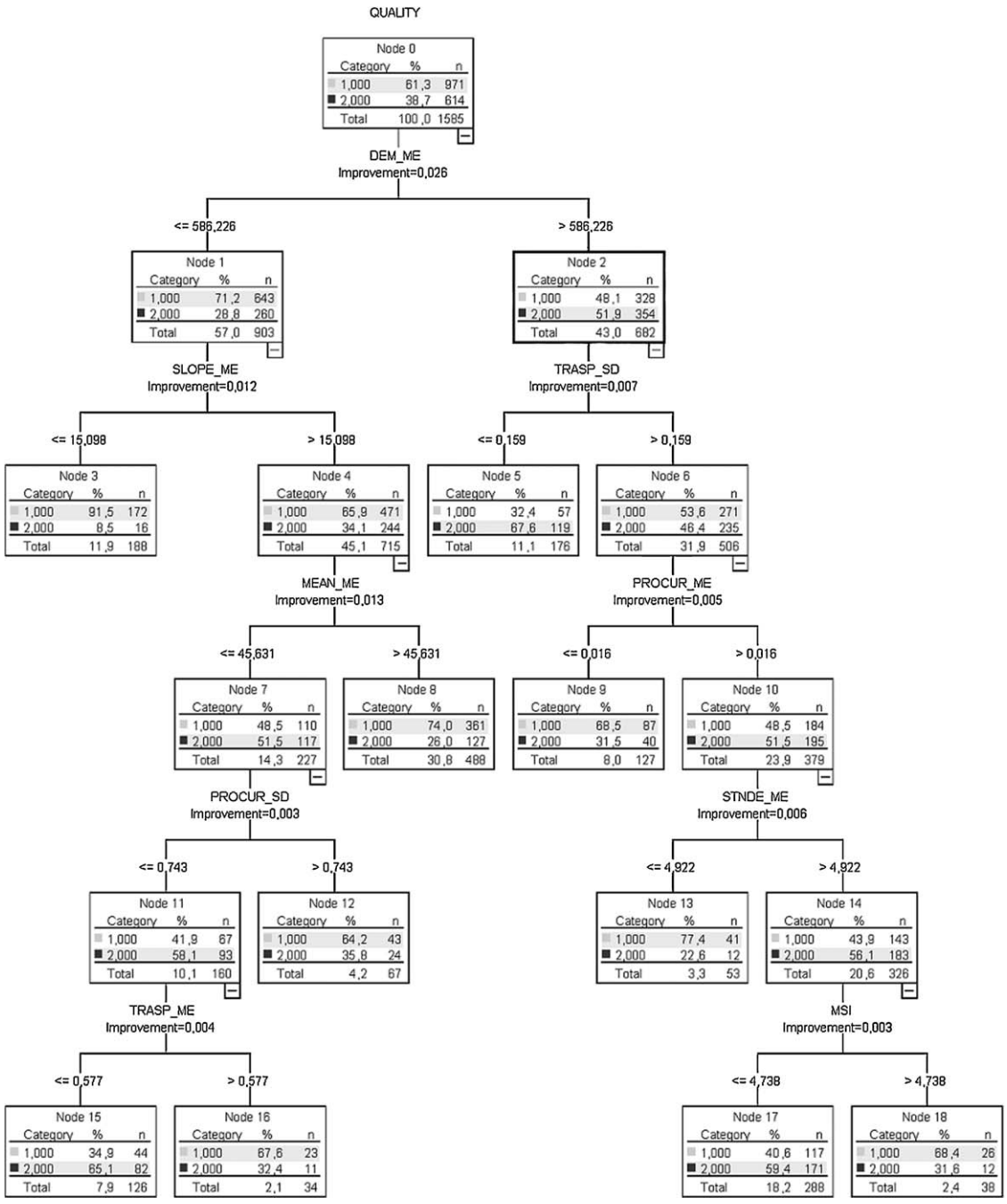


Fig. 3. Classification tree.

and more convoluted patches (MSI > 4.7) defined a group of high quality stands.

4. Discussion

Chestnut forests occur in a variety of environments in the study area, ranging from Atlantic to Mediterranean climates, across a wide range of altitudes and terrain relief forms (Gandullo Gutierrez et al., 2004; Rubio et al., 2000, 2002; Vieitez Cortizo et al., 1996) and are subject to different land use dynamics and management (Álvarez et al., 2010; Bouhier, 1979; Díaz Hernández et al., 2009; Fernández López, 1984). As a consequence, stand structure tends to be highly heterogeneous across the area (Gandullo Gutierrez et al., 2004; Rubio et al., 2000, 2002).

Despite the heterogeneity of conditions and stand characteristics, the results indicated strong relationships between chestnut stand quality (based on maturity and purity of the stand) and the concurrence of certain types of terrain relief, patch geometry and spectral response. The model also performed well, even though we did not include socioeconomic variables, which may have played a major role in discriminating between types of stand structure and quality. As a plausible explanation, some terrain attributes related to accessibility (e.g. slope) may have operated as surrogates for socioeconomic variables related to the management intensity of the stands (Zerger et al., 2009).

The classification tree obtained combines a total of nine predictors that correctly assigned the quality class in more than 70% of

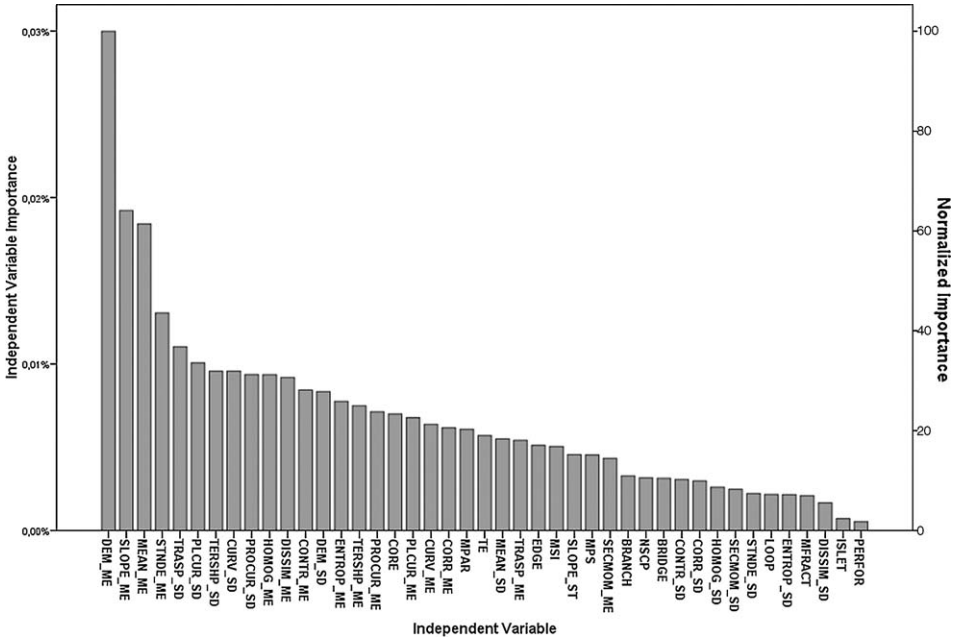


Fig. 4. Relative and normalised importance of the predictors (independent variables) in the CaRT.

cases, and also showed fairly good agreement between observed and predicted qualities, according to the values of the Kappa statistic. However, terminal nodes of the classification tree tended to show a certain degree of impurity, particularly in the groups classified as class 2 (good quality stands). This is consistent with the tendency for seminatural and managed habitats to be prone to stochastic controls in their distribution, and the difficulty in them being described by deterministic models (Zerger et al., 2009). In addition, in particular cases, the geographic distribution of chestnut – like other cultivated and managed species – may be at the limits of its ecological range (Conedera et al., 2004b), which may also have hindered assessment of environmental controls.

The application of CaRT enabled us to identify the most relevant predictors of chestnut stand quality, and also to model interactions and synergistic effects (Elith et al., 2006). Hence, by using the classification tree pathway, we were able to trace some major patterns in the conservation status of chestnut stands in relation to topographic, morphological and spectral variables.

In fact, CaRT enabled us to identify terrain attributes as the main discriminating factors in the distribution of chestnut stands. This may have been due to the stronger influence of geomorphological and climatic factors on ecological patterns at broad scales (Meisel and Turner, 1998; Swanson et al., 1988), such as the scale of analysis in the present study.

In the study area, chestnut stands rarely occur at altitudes higher than 1100–1200 m because of climatic constraints (Berrocal del Brío et al., 2006). The present results confirmed that altitude plays a major role in the geographic distribution of the species, and also in the distribution of stand quality, as stated in previous studies carried out at a lower geographic scale (Fernández-Manso et al., 2010). Hence, good quality (pure and/or mature) chestnut stands tend to occur at altitudes above 600 m, which in the study area roughly marks the difference between lowlands and highlands (Ramil Rego et al., 2005), which is also consistent with the co-occurrence of a series of conditions that may historically have favoured the development and persistence of chestnut stands. Indeed, analysis of historical records of chestnut distribution in the region of Galicia have revealed serious ink disease infections (*Phytophthora cambivora* and *P. cinnamomi*), since the 19th century, in the lowlands located along the coastal fringe, and subsequent retraction of chestnut stands towards inland areas at relatively high altitudes (Bouhier, 1979). In addition, the introduction of corn in the 17th century, along with the spread of *Pinus pinaster* in the 18th century and more recently *Pinus radiata* and *Eucalyptus* spp., have probably taken over a large area originally devoted to chestnut stands in the lowlands (Bouhier, 1979).

Despite this historical background, it is possible to find good quality chestnut stands at altitudes lower than 600 m, providing that certain conditions related to terrain relief and spectral texture are first met. Firstly, the slope must be steeper than 15 degrees, a value that could be interpreted as indicating a low interest of these locations for intensive agricultural use and also less vulnerability to fungi diseases because of better drainage (Erwin and Ribeiro, 1996; Mansilla Vázquez et al., 2000). The second predictor indicates that at this range of altitude and slope, stands with low average NDVI (predictor MEAN-ME) are more likely to correspond to high quality chestnut stands. This texture feature indicates that for the spatial domain used (a 5×5 pixel window, corresponding to 2.25 ha), the overall tone is somehow dominant against spatial differences in spectral response, and also that this type of stand does not obtain particularly high values of greenness NDVI index ($\leq$45 out of a distribution ranging from 10 to 62). In short, chestnut stands with low local variance in their spectral response and relatively low values of greenness are more likely to be high quality stands than younger and/or mixed chestnut stands, which have heterogeneous and green canopies. In this case, this feature may indicate either a low effect of the spatial domain in the spectral response of the high quality stands or simply that the image resolution or the window dimensions were not able to capture local variations in the texture. In fact, previous studies used finer image resolutions to discriminate forest stand structure (Franklin et al., 2001; Kayitakire et al., 2006; Wunderle et al., 2007), whereas in other cases medium resolution imagery such as Landsat TM/ETM+ or SPOT4 was sufficient for discrimination of forest stand characteristics (Cohen and Goward, 2004; Hansen et al., 2001) depending on crown dimensions and density. Therefore, it may be advisable in future research to test texture features from finer resolution images, as some potential texture predictors may have been rejected by the model due to

a lack of quality/resolution rather than a lack of causal association (Zerger et al., 2009).

Secondly, high quality stands are associated with the concurrence of low variation in profile curvature (i.e. in the rate of change in slope) and cold and wet exposure. The second predictor is easily interpreted as the fidelity of the best stands for northern aspects in the lowlands, in agreement with the general species preference in the NW Iberian Peninsula (Rubio and Gandullo, 1994; Rubio et al., 2000), despite some other considerations (discussed further below). The importance of exposure is also highlighted by the relatively steep slope marked by previous splits in the decision tree. The relationship between high quality stands and uniform along-slope profile is not straightforward. In general terms, we can state that stand quality is sensitive to this variable, as patches with high standard deviation in the parameter tend to correspond to low quality. This may be attributed to the fact that patches with rough terrain are more likely to show variations in micro-environments and therefore to include species or stand structure mixtures. Besides, the study area covers a large variety of environments, ranging from Atlantic to Continental and Mediterranean areas, and chestnut probably develops better in convex topographies in wetter climates, while it benefits from the higher water budget of concave slopes in dryer environments (Masson, 2005). Therefore, even when it was not possible to assign quality groups to certain profile curvature values, we were able to discriminate quality groups according to their fidelity to one or other type of homogeneous curvature.

In mountain settings (above 600 m a.s.l.), it was not possible to detect a clear relationship between a given value of aspect and chestnut stand qualities. Nevertheless, as the standard deviation of aspect (TRASP_SD) arose as the first split criterion in this branch, chestnut stand quality is clearly sensitive to this parameter. Previous studies on chestnut autoecology in the area showed differences in species behaviour in relation to terrain morphology, depending on the particular study area, methodology and/or data sources. Hence, some studies concluded that chestnut stands are more frequent under northern aspects or locations with low insolation (Rubio et al., 2000, 2002), while other studies suggest low sensitivity of the species to aspect/insolation (Gandullo Gutierrez et al., 2004), and even a preference for sun-facing slopes under wet and cold climates (Berrocal del Brío et al., 2006). In accordance with the present results for altitudes higher than 600 m, good quality stands are located on homogeneous slopes with a dominant aspect, which may vary throughout the study area depending on particular locations. In other words, good quality stands may face different directions depending on regional scale environmental controls: i.e. south in Oceanic environments and north in Mediterranean settings, although this finding must be confirmed in further studies.

The remaining group of good quality stands is defined by the concurrence of concave slopes, relative heterogeneous spectral response (textured patches) and relatively compact patch morphology. In contrast with lowland chestnut stands, in this case there is a relation between concave terrain morphology and good stand quality, as chestnut stands on exposed settings tend to correspond to young and/or mixed forests. This evidence contrasts with the theoretical preference of the species for exposed positions, in order to avoid waterlogging and "frost pockets" (Masson, 2005) and also with the higher risk of uprooting tall trees in such locations due to faster growth (Vogt et al., 2006). In any case, the dominance of mixed or young chestnut stands on convex sections of slope may correspond to recent colonisations on optimal environments for the species, for example after a period without any fire. In addition, the effect of concave topography in these locations is compensated by altitude and colder climate in relation to susceptibility to fungal diseases, making the presence of good quality chestnut stands possible. Moreover, this form of relief may act as a shelter against certain type of disturbance, such as severe fire.

Spectral texture also played an important role in discriminating stand quality. In this case, high quality stands displayed a more textured spectral response (at the scale considered) than low quality stands. This may be related to the heterogeneity of the crown canopy, due to the relatively large size of the individuals in mature stands, which was captured by a window size of 5×5 pixels in an image of 30 m of spatial resolution. Low quality stands corresponded to a finer texture, probably because of the smaller size of crowns and higher densities, despite the higher probability of species mixtures.

The results also showed that, along with the concurrence of other predictor values, low patch irregularity enables prediction of high chestnut stand quality. Indeed, following the CaRT pathway, good quality chestnut patches in the highlands occur on slopes with uniform aspect and concave topography, and also with a heterogeneous spectral response of canopy cover and a relatively compact shape.

There are many examples illustrating the use of vegetation patch morphology to predict human alteration of the landscape (Moser et al., 2002) and particularly in forest patches (Saura and Carballal, 2004). In the present case, the representativeness of spatial pattern predictors in the CaRT analysis was limited to Mean Shape Index (MSI), enabling us to determine a relationship between stand quality and patch compactness. In fact, this landscape metric made it possible to discriminate a group of high quality stands for MSI values lower than 4.7, and therefore more compact shapes than low quality stands. Indeed, even when absolute MSI values are difficult to interpret, the results show that at this terminal node of the CaRT, low quality stands correspond to a rather convoluted shape. This is consistent with the findings of Saura and Carballal (2004), which confirmed the tendency for monospecific native forest patches to be more compact than mixed native forests. In addition, traditional chestnut forest and orchards, as semi-natural and managed habitats, usually lead to more or less complex land use systems. In such systems, compact and regular patch edges are frequent or even dominant in the landscape, as a result of human disturbance or management (Moser et al., 2002). This may also explain the relationship between compact patch shapes and good quality chestnut stands.

5. Conclusions

In the present study, we explored the capacity of terrain, spectral and landscape configuration descriptors to predict the conservation status of chestnut stands in NW Spain, defined by a binary quality index that combines stand purity and maturity.

The use of classification and regression trees enabled identification of several factors related to geomorphological features, heterogeneity of spectral response in remote-sensed data and patch morphology as potential predictors of chestnut conservation status. We also identified and interpreted the overall patterns in the relations between the predictor values and the chestnut stand condition in terms of the interaction between environmental conditions and historical human activity.

Altitude emerged as a major indicator of the quality of chestnut forest stands, enabling us to discriminate two clear spatial domains identified as lowlands and highlands. Different combinations of terrain attributes, remote sensed image texture features and patch shape compactness enabled discrimination of different groups of chestnut stands according to the quality index. The results show that good quality chestnut patches only occur under particular combinations of terrain characteristics and show relatively smooth spectral texture in the lowlands in the study area. Chestnut stands

may therefore occur in the highlands under a wider range of conditions, as long as the slope shows low variation in aspect, or a certain combination of terrain, canopy spectral response and patch shape occurs.

Identification of the main elements determining chestnut forest quality is essential for the management of a threatened Natura 2000 habitat of interest and cultural landscape element, and of a complete land use system now facing an important process of abandonment. The design of management strategies to maintain chestnut stands in the framework of new land use dynamics may benefit from this contribution to the identification of key factors linked to the location of well conserved chestnut stands.

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